Automated Exoplanet Detection Pipeline

Hunting for Earth-sized worlds in TESS Photometric Data using **Deep Transfer Learning**, **Synthetic Augmentation**, and **SNR Gating**.

The Problem

- **The Terrestrial Deficit:** Baseline models miss faint Earth-sized planets (Recall: 31.2%).
- **Mass-Radius Degeneracy:** Photometry alone cannot distinguish between giant planets and small brown dwarfs.
- **The Hallucination Effect:** Hyper-sensitizing AI to find faint U-shapes causes it to hallucinate planets in pure static.

The Solution

- **Synthetic Augmentation:** Injecting artificial low-SNR transits during training to teach the AI to see through noise.
- **Tri-Input Architecture:** Fusing Global Views, Local Views, and physical Stellar Metadata (Radius/Mass).
- **Binned SNR Gatekeeper:** A mathematically forgiving 2.0σ filter to block egregious static before AI inference.

The Autonomous Pipeline

1. Raw TESS Data
2. BLS Algorithm — Period & Epoch Search
3. SNR Gatekeeper — Filters if SNR < 2.0
4. Tri-Input CNN — Deep Classification
5. Planet Candidate

Empirical Metrics (Validation Set)

BASELINE MODEL

31.2%

Planetary Recall

Conservative AI missed faint Earth-analogs. (Precision: 85.5%)

AUGMENTED MODEL

91.9%

Planetary Recall

Synthetic injection tripled sensitivity but caused hallucinations. (Precision: 59.3%)

HYBRID + GATEKEEPER

90.3%

Preserved Recall

SNR Gate successfully restored precision (60.1%) while keeping high recall.

Cross-Mission Validations

TOI-700 d — SUCCESS

Habitable Earth-Analog

Binned SNR:	2.41
AI Confidence:	99.99%

The ultimate test. Forgiving gatekeeper preserved this faint signal, proving the pipeline works on real terrestrial data.

TOI-125 b — SUCCESS

Sub-Neptune

Binned SNR:	2.42
AI Confidence:	99.83%

Would have been deleted by a standard 3.0σ filter. The Tri-Input CNN correctly identified the morphology.

TIC 105706043 — EDGE CASE

Instrumental Static

Binned SNR:	2.69
AI Confidence:	100.00%

Perfectly timed jitter spoofed the mathematical gatekeeper, triggering a hallucination. Highlights need for cloud-scale noise modeling.